

Midterm project

Nikhil Kumar

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1 Introduction

The Remaining Useful Life (RUL) of the mechanical systems is a key issue, which needs to be predicted in the context of predictive maintenance. Unexpected engine failures may cause a great deal of financial losses and even serious safety risks in such industries as aviation. Conventional maintenance plans such as planned and reactive maintenance are not usually efficient since they either prematurely replace parts or maintain them without stopping failures. Consequently, more and more interest in data-driven methods that can approximate system health and anticipate failures before they happen is growing.

The project at hand is concerned with the prediction of Remaining Useful Life of jet engines based on NASA CMAPSS (Commercial Modular Aero-Propulsion System Simulation) data [1]. This data has multivariate time-series data of various engines at different conditions up to failure. At each time step, operational settings and several sensor measurements are available and the goal is to make a forecast of the number of cycles to go before an engine fails.

This is a regression time-series problem, and the model has to acquire degradation tendencies over time. To find a solution to this, we apply baseline machine learning models and a deep learning methodology. In particular, we compare linear regression and random forest regression to a 1D Convolutional Neural Network (CNN) that is implemented to extract the temporal relationships in sequential data. This project aims at assessing the performance of these models and find out whether deep learning techniques can enhance the accuracy of prediction when estimating the RUL.

2 Related Work

The prediction of Remaining Useful Life (RUL) is a predictive maintenance technique that has been extensively studied and referenced in relation to its importance in ensuring that systems are made more reliable and that operation costs are minimized. Degradation is a characteristic of complex systems like aircraft engines that is affected by changes in different sensor measurements; therefore, degradation is more applicable to data-driven approaches.

As a basic approach to RUL prediction, traditional machine learning approaches have been used, which include linear regression and tree-based approaches. However, these approaches are deemed ineffective in capturing time-based dependencies that are characteristic of sequence-based data. In addition, to address this limitation of traditional approaches, recent studies have been conducted on deep learning approaches, particularly those that are based on sequence-based approaches like RNN and LSTM [2].

More recently, time-series prediction tasks (and RUL estimation) have also been performed by using Convolutional Neural Networks (CNNs) [3]. CNNs can capture local temporal features and have few parameters than recurrent ones which is computationally efficient and yet capable of obtaining good performance.

In the given project, we feed a 1D CNN model with the engine sensor data to identify time-related trends in the data and compare its functionality to other baseline models, including linear regression, and random forest regression. This will enable us to assess the claim that deep learning methods have a quantifiable benefit in predicting RUL on the NASA CMAPSS dataset.

3 Methods

3.1 Dataset and Preprocessing

We utilize a benchmark dataset named NASA CMAPSS FD001. This dataset is a multivariate time series dataset that contains information collected from multiple jet engines running under a single condition with one fault mode. Each row in this dataset corresponds to a single engine cycle. It contains information about an engine’s identifier, cycle number, three operational setting values, and twenty-one sensor values.

The training dataset has complete run-to-failure trajectories for every single engine. This enables us to calculate the remaining useful life. In this dataset, we calculate the remaining useful life for every single time step. This is done by subtracting the current cycle number from the maximum cycle number for every single engine.

In the pre-processing step, we loaded the raw data files. This is done using a whitespace-separated parsing mechanism. Trailing empty columns were also removed. Every single column is given a specific name corresponding to every single engine’s identifier, cycle number, operational setting values, and sensor values.

3.2 Feature Engineering

To build the input features, the engine ID and cycle number were removed to prevent the use of identification variables or absolute time as a predictor. The remaining operational variables and sensor readings were used as the features.

Standardization of the numeric features was performed by applying a standard scaling technique.

3.3 Models

Two baseline regression models were implemented for comparison:

- **Linear Regression:** A simple model that assumes a linear relationship between input features and the target variable.
- **Random Forest Regression:** An ensemble-based model that captures nonlinear relationships and interactions between features.

To prevent data leakage across time steps coming from the same engine, an engine-level train validation split is used. This ensures that all cycles from a given engine occur within only one split.

3.4 Evaluation Metrics

Model performance was evaluated using two standard regression metrics:

- **Mean Absolute Error (MAE):** Measures the average absolute difference between predicted and true RUL values.
- **Root Mean Squared Error (RMSE):** Measures the square root of the average squared differences, placing greater emphasis on larger errors.

4 Preliminary Results

Preliminary experiments were performed on the NASA CMAPSS FD001 dataset to assess baseline performance on the Remaining Useful Life (RUL) prediction task. This dataset has 20,631 training samples and 13,096 test samples, each of which is associated with 100 engines.

Two baseline models, Linear Regression and Random Forest Regression, were trained on standardized features from sensors and operations. For evaluation, a validation split was used on an engine level to prevent leakage.

The results are given below:

- **Random Forest Regression:** MAE = 25.874, RMSE = 35.400
- **Linear Regression:** MAE = 30.111, RMSE = 38.133

The results show the superiority of the Random Forest model over Linear Regression, which confirms the importance of nonlinear relationships in the sensor data for the prediction of RUL. However, the results show the limitation of both models in handling the temporal degradation patterns in the data.

The above results show the importance of using sequence-based models, such as Convolutional Neural Networks (CNNs), which are more efficient in handling the temporal dependencies in the engine degradation data. Further work will be done in this direction to improve the prediction accuracy.

5 Future Plan

On the basis of these preliminary results, further research will be conducted to enhance the performance of the models by incorporating techniques for temporal modeling. For this purpose, sequence-based models, such as Convolutional Neural Networks (CNN), will be incorporated for better degradation pattern recognition. Further improvements in this research will be made by trying out different window sizes for sequence generation, tuning the parameters of the models, and exploring feature selection techniques for identifying the best sensors. Further evaluation of the models will be conducted by employing both MAE and RMSE for assessing the models' performance against different error measures. Moreover, a comparative study of baseline models and deep learning models will be conducted to evaluate the effectiveness of temporal models in enhancing the accuracy of predicted Remaining Useful Life.

6 References

References

- [1] A. Saxena, K. Goebel, D. Simon, and N. Eklund, "Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation," International Conference on Prognostics and Health Management, 2008.
- [2] S. Zheng, K. Ristovski, A. Farahat, and C. Gupta, "Long Short-Term Memory Network for Remaining Useful Life Estimation," IEEE International Conference on Prognostics and Health Management, 2017.
- [3] X. Li, Q. Ding, and J. Sun, "Remaining Useful Life Estimation in Prognostics Using Deep Convolution Neural Networks," Reliability Engineering & System Safety, 2018.